

Atharva Tendle

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SUMMARY

Senior Research Engineer with a robust background in generative AI, large language models, and AI safety, with hands-on experience leading AI projects from conception to production. Strong expertise in model architecture, algorithm design, and large-scale data processing. Demonstrated ability to balance cutting-edge research with practical engineering to build high-performance AI systems. Passionate about AI ethics and safety, and experienced in improving model interpretability and monitoring.

EXPERIENCE

Senior Research Engineer - AI

06-28-2021 – Present

Bloomberg L.P. (AI - Group)

New York City

- **Gen AI Retrieval & Agentic Frameworks:** Building a generalizable framework for retrieval and agentic workflows focusing on Indexing, Retrieval, Evaluations and Agentic Systems.
- **Generative AI Large Language Models:** Led development of news summarization systems using LLama3.1 70B models, integrating post-processing tools for AI safety, improving content reliability, and enabling human-in-the-loop review.
- **Research Prototyping:** Designed and prototyped LLM-driven Entity Linking, enhancing NER, NED and Entity Salience tasks through innovative techniques like 0/1-shot prompting, CoT, and context augmentation, resulting in significant F1-score improvements.
- **Optimizing ML Infrastructure:** Identified bottlenecks and improved model efficiency both on a software and ML-level resulting in a $\sim 50\%$ drop in latency during inference.
- **Model Interpretability Monitoring:** Developed interpretability tools for real-time model monitoring and error analysis, improving pipeline reliability through diagnostic UIs and online error tracking systems.

Deep Learning Research Assistant

12-01-2019 – 05-07-2021

Network-centric and data-driven learning group (UNL)

Lincoln, NE

- Led research on self-supervised learning, achieving state-of-the-art results on camera-trap datasets.
- Reduced training time by 50% while improving classification performance 93.5% – 94.5%.

Data Science Intern

05-11-2020 – 08-07-2020

Hudl: Applied Machine Learning

Lincoln, NE

- Worked on Multi-Object Tracking using state-of-the-art Deep Learning techniques like Detectron2 and DeepSORT
- Improved object detection models using multi-object tracking techniques, boosting detection performance to 92% through optimized training pipelines and large-scale training setups.

THESIS

“The Revolution Will Not Be Supervised”: An Investigation of the Efficacy and Reasoning Process of Self-Supervised Representations

JOURNAL ARTICLES

- Atharva Tendle, Mohammad Rashedul Hasan. *A study of the generalizability of self-supervised representations.* Journal of Machine Learning with Applications, Elsevier, 2021

CONFERENCE PAPERS

- Rajarshi Bhowmik, Marco Ponza, **Atharva Tendle**, Anant Gupta, Rebecca Jiang, Xingyu Lu, Qian Zhao, Daniel Preotiuc-Pietro, *Leveraging Contextual Information for Effective Entity Salience Detection*, In Findings of the Association for Computational Linguistics: NAACL 2024, pages 395–408, Mexico City, Mexico. Association for Computational Linguistics.
- **Atharva Tendle**, Andrew Little, Stephen Scott, and Mohammad Rashedul Hasan, *Self Supervised Learning in the Twilight of Noisy Real-World Datasets*. 21st IEEE International Conference on Machine Learning and Applications (ICMLA) 2022.

- Ayush Manish Agrawal, **Atharva A. Tendle**, Harshvardhan D. Sikka, Sahib Singh, and Amr Kayid. *Investigating Learning in Deep Neural Networks using Layer-Wise Weight Change*. Computing Conference 2021
- Ayush Manish Agrawal, **Atharva A. Tendle**, Harshvardhan D. Sikka, Sahib Singh, and Amr Kayid. *Investigating Learning in Deep Neural Networks using Layer-Wise Weight Change*. In Advances in Neural Information Processing Systems(NeurIPS) NewInML Workshop 2020
- Harshvardhan D. Sikka, **Atharva A. Tendle**, and Amr Kayid. *Multimodal Modular Meta-Learning*. In Advances in Neural Information Processing Systems (NeurIPS) Preregister Workshop 2020
- D.Newton , D.Piatkowski , W.Marshall , **A.Tendle**, *Deep Learning Methods for Urban Analysis and Health Estimation of Obesity*, eCAADe 2020

TECHNICAL SKILLS

Frameworks: LlamaIndex, Langchain, PyTorch, Pytorch-Lightning, HuggingFace, KServe, Kafka

Developer Tools: Git, AWS S3, VS Code, Jupyter, Docker

Libraries: Pandas, NumPy, Matplotlib, Scikit-Learn, OpenCV

EDUCATION

University of Nebraska-Lincoln

Lincoln, NE

Master's of Science in Computer Science GPA: 4.0/4.0

Aug. 2019 – May 2021

University of Nebraska-Lincoln

Lincoln, NE

Bachelor's of Science in Computer Science & Economics GPA: 3.91/4.0

Aug. 2015 – May 2019